

Integrative Neuromuscular Training and Injury Prevention in Youth Athletes. Part I: Identifying Risk Factors

Azahara Fort-Vanmeerhaeghe, PhD,^{1,2,3} Daniel Romero-Rodriguez, PhD,¹ Alicia M. Montalvo, MS, LAT, ATC, CSCS,⁴ Adam W. Kiefer, PhD,^{5,6,7} Rhodri S. Lloyd, PhD, CSCS*⁸, and Gregory D. Myer, PhD, CSCS*^{5,6,9,10}

¹School of Health and Sport Sciences (EUSES) Universitat de Girona, Salt, Catalonia, Spain; ²Blanquerna Faculty of Psychology, Education Sciences and Sport (FPCEE), Universitat Ramon Llull, Barcelona, Spain; ³Segle XXI Female Basketball Team, Catalan Federation of Basketball, Esplugues de Llobregat, Spain; ⁴Department of Athletic Training, Nicole Wertheim College of Nursing and Health Sciences, Florida International University, Miami, Florida; and Department of Kinesiology, Athletic Training/Sports Medicine Program, Pennsylvania State University, University Park, Pennsylvania; ⁵Division of Sports Medicine, Cincinnati Children's Hospital Medical Center, Cincinnati, Ohio; ⁶Department of Pediatrics, College of Medicine, University of Cincinnati, Cincinnati, Ohio; ⁷Department of Psychology, Center for Cognition, Action and Perception, University of Cincinnati, Cincinnati, Ohio; ⁸Youth Physical Development Unit, Cardiff Metropolitan University, Cardiff, Wales; ⁹Sports Health and Performance Institute, Ohio State University, Sports Medicine, Ohio State University Medical Center, Columbus, Ohio; and ¹⁰Micheli Center for Sports Injury Prevention, Waltham, Massachusetts

ABSTRACT

PART I OF THIS REVIEW DESCRIBES THE MOST IMPORTANT NEUROMUSCULAR SPORTS INJURY RISK FACTORS IN YOUTH ATHLETES: MUSCLE FATIGUE, ALTERED TIMING AND MAGNITUDE OF MUSCLE ACTIVATION, STRENGTH DEFICITS, PREDOMINANCE OF FRONTAL PLANE CONTROL STRATEGIES, NEUROMUSCULAR IMBALANCES BETWEEN LIMBS, INADEQUATE MUSCLE STIFFNESS, DEFICITS IN POSTURAL STABILITY, ALTERED PROPRIOCEPTION, AND FEED-FORWARD CONTROL. THE SECOND PART OF THIS REVIEW PROVIDES A FLEXIBLE APPROACH

TO INTEGRATIVE NEUROMUSCULAR TRAINING WITH THE GOAL TO IMPROVE INJURY RESILIENCE AND TO ENHANCE SPORT AND MOTOR SKILL PERFORMANCE.

INJURIES IN YOUTH ATHLETES

Recent data indicate that approximately 8.2 million youths between the ages 6–12 participate in organized sport in the United States (112). However, recent data indicate that participation in organized sports activities does not ensure that youth meet physical activity (PA) recommendations (59,93). This high participation in youth sports has an inherent risk of injury (4). Many injuries in youth result from traumatic incidents; however, 30–50% of injuries are estimated to result from overuse during sport participation (96). Thus, it may be important to emphasize the

correct development and training of sport skills to improve participation outcomes in youth, as the development and execution of incorrect motor patterns during sports activities may increase the potential risk of sport injury (74).

Relative to adults, the rate of sport injury in youth athletes is lower (10), but consequences are more adverse; therefore, sports injuries in youth can become a barrier to long-term PA throughout the lifespan (15,24). Inadequate PA, in turn, is associated with high morbidity and long-term disability (112,119), and injuries that disrupt PA represent a considerable social and economic burden (19). Furthermore,

KEY WORDS:

high-risk biomechanics; pediatrics; youth sports

Address correspondence to Azahara Fort-Vanmeerhaeghe, afortvan@gmail.com.

previous injury in youth may result in a loss of enthusiasm for PA stemming from fear of injury or reinjury (10,29). For example, an investigation into the risk factors associated with injuries that resulted from physical education, leisure time PA, and sports participation in 9–12 year olds indicated that gender, age, and, most importantly, previous level of PA were all significantly related to injury. Thus, the researchers concluded that PA promotion efforts should include a focus on injury prevention (10).

Given the negative effects of injury in youth both acutely and throughout the lifespan, injury prevention strategies should be implemented early to avoid negative consequences. Sports injuries in youth athletes have been associated with growth and development (48,85), low levels of physical fitness (10), inadequate physical preparation (56), diminished motor abilities (1), and deficits in fundamental movement skills (77). Therefore, optimal levels of physical conditioning and neuromuscular coordination are important factors to address in youth athletes, especially in situational or team sports (89,91,110). Moreover, some of the most modifiable risk factors in youth athletes are abnormal movement patterns (e.g., dynamic valgus of the knee during landing maneuvers) during execution of sport skills (81). Deviations from desirable movement patterns are usually associated with deficits in neuromuscular control strategies (Figure 1). Current research indicates that neuromuscular risk factors associated with sports-related injuries can be modified through effective integrative neuromuscular training (INT) programs to directly reduce the incidence of injury in young athletes (6,25,78). INT entails a training program that incorporates general and specific strength and conditioning tasks with the goals of improving injury resilience and enhancing sport and motor performance skills (77).

Although there is no consensus regarding optimal neuromuscular training programs, evidence supports the use

of such programs to prevent injury (113–116). This review of neuromuscular risk factors in youth athletes may provide a flexible approach to optimize the design of INT programs. This article is Part I of a two-part review. Part I aims to synthesize the latest literature with regard to neuromuscular risk factors for injury to optimize targeted INT programs in the youth sports population. Part II aims to provide practical applications for coaches to implement INT to reduce injury risk and improve performance in the youth sports population.

INTEGRATIVE NEUROMUSCULAR TRAINING TO REDUCE RISK OF INJURY IN YOUTH ATHLETES

Research indicates that neuromuscular and biomechanical risk factors can be modified by INT programs (52,82,84,90). These programs have been shown to directly reduce the incidence of injury in athletes (41,52). In addition, there is an established age-related association between INT program implementation and reduction in anterior cruciate ligament (ACL) injuries (86). This association indicates that the effects of puberty on the neuromuscular system can be mitigated. Puberty is characterized by major musculoskeletal developments that usually are not accompanied with sufficient corresponding neuromuscular adaptation, and this may result in the development of abnormal mechanics during sport actions (49,103,121). Consequently, it may be optimal to initiate INT programs in the prepubescent athlete to establish movement patterns that can limit the development of motor deficits that may result from puberty.

Neuromuscular control during sport actions depends on the proper functioning of the sensorimotor system. This complex system incorporates afferent and efferent nervous system signals, as well as central integration and processing components involved in maintaining dynamic joint stability (104). Although definitions vary (49,103), this review will operationally define neuromuscular control as the precise muscle activation that allows

for a coordinated and efficient action (37). Figure 2 provides a schema depicting the relationship between training, the sensorimotor system, joint stability, and neuromuscular control, and the effect of each on injury risk. An improvement in the sensorimotor system through task-specific training leads to increased neuromuscular control, which can then improve dynamic joint stability during intense sports maneuvers. The concomitant improvements in neuromuscular control and dynamic joint stability can decrease the risk of injury.

ANALYSES OF NEUROMUSCULAR INJURY RISK FACTORS IN YOUTH ATHLETES

As previously mentioned, the optimization of INT programs in youth athletes arises from the knowledge of mechanisms by which neuromuscular risk factors underlie subsequent sports injuries. This review will focus on the most commonly injured areas of the body in youth athletes: the ankle and the knee (1). The current literature emphasizes the following neuromuscular risk factors of injury (36): muscle fatigue (11,67), altered timing and magnitude of muscle activation (31,70,126), strength deficits (38,42,100), predominance of frontal plane control strategies (dynamic valgus) (5,87), neuromuscular imbalances between limbs (16,76), inadequate muscle stiffness (39), deficits in postural stability (99), altered proprioception (105), and feed-forward control (9) (Figure 1). Table 1 provides an overview of INT strategies that can be used to target neuromuscular risk factors in youth athletes.

MUSCLE FATIGUE

Muscle fatigue is defined as a progressive decrease of maximal force and power capacity of muscles, which means that submaximal contractions can be sustained after the onset of muscle fatigue (26). Fatigue is associated with changes in neuromuscular control strategies during sport tasks and decreases in dynamic joint stability in lower limbs (11,18,67). Moreover, fatigue is

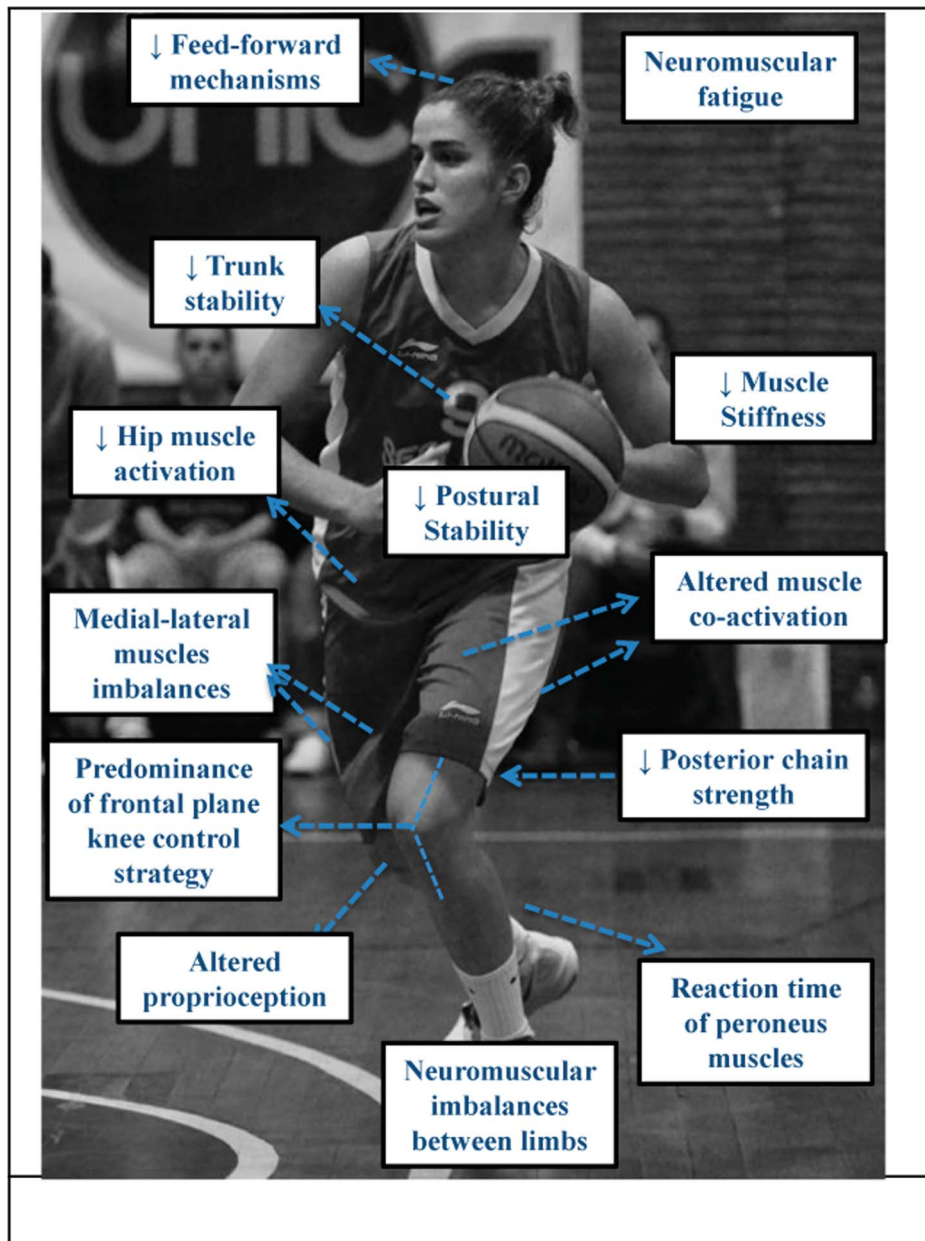


Figure 1. Relationship between altered kinematics and neuromuscular risk factors (adapted with permission from Fort-Vanmeerhaeghe and Romero-Rodriguez, 2013).

associated with reduced coordination, altered proprioception, and changes in lower limb biomechanics, such as decreased knee and hip flexion, increased dynamic valgus at the knee, increased ground reaction forces, and increased time to joint stabilization (12,17,33,109). Decreased neuromuscular control resulting from fatigue is associated with

an increase in risk factors for sports injuries such as ACL ruptures (12,17) and ankle sprains (33). Importantly, different sexes respond to neuromuscular fatigue in different ways (55). Female athletes suffer greater consequences related to injuries compared with male athletes (46), so sex is important to consider when creating INT programs. High-intensity

training of neuromuscular tasks is likely important, especially under conditions of fatigue, to mimic sport conditions and train the neuromuscular system to perform in a safe and efficient manner while fatigued. One final consideration for training youth athletes is their capacity to resist muscle fatigue and their ability to recover during high-intensity

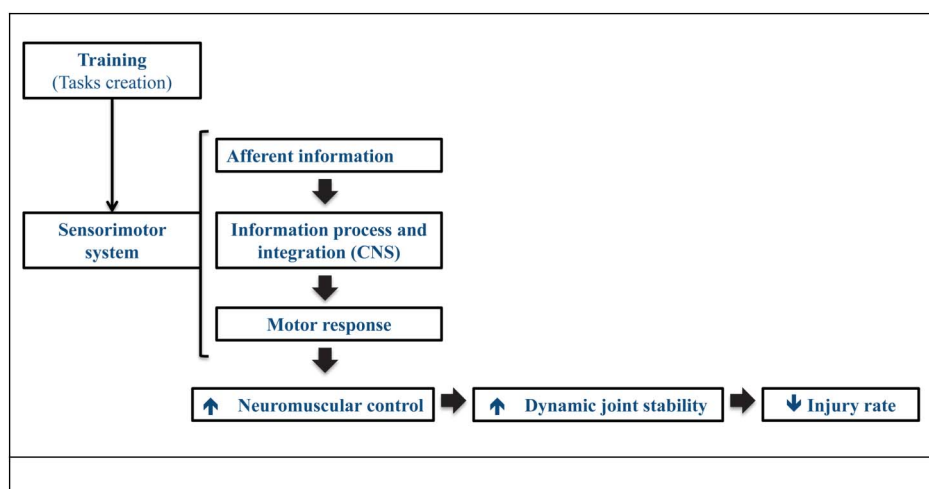


Figure 2. Relationship between training, sensorimotor system, joint stability, and neuromuscular control for prevention of sports injuries in youth athletes (adapted with permission from Fort-Vanmeerhaeghe and Romero-Rodriguez, 2013).

intermittent exercise (23,101). Pre-pubescent children are able to resist acute fatigue better than adolescents and adults during one or several repeated high-intensity exercise bouts (101). Although the exact mechanism is not known, it is possible that because children have less muscle mass than adults, they generate lower absolute power during high-intensity exercise. In addition, there is a significant interaction between maturity and sex on fatigue resistance during high-intensity intermittent exercise after the onset of puberty. Adolescent males have greater fatigue resistance than adult males, whereas no differences between adolescent females (mid puberty 14–15 years) and adult females (23) have been found. Therefore, youth may be able to perform with greater levels of neuromuscular control when in a fatigued state compared with adults in an INT program. However, youth may need longer recovery periods than adults to ensure that adequate motor control strategies are used during training in high fatigue conditions. It is also important to note that, while children can resist acute fatigue to a greater extent compared with adults, we do not yet fully understand the

potential risks of accumulated fatigue in youth.

ALTERED TIMING AND MAGNITUDE OF MUSCLE ACTIVATION

Alterations in muscle activation capacity during sport-specific tasks may increase the risk of injury (20,42,70). Alterations in muscle activation capacity refer to the changes in intensity of muscle activation and in time to reach the maximal activation. Investigations carried out mainly using electromyography indicate the following associations between injury risk factors and specific injuries:

Delay in reaction time of peroneal muscles. Although there is a lack of consensus in the scientific literature, functional ankle instability (FAI) has been associated with increased peroneal reaction time (61,70,71). A recent review evaluated whether peroneal reaction time was influenced relative to an injured or previously injured ankle (70). The authors showed a significant delay in the reaction time of peroneal muscles in the injured ankles. Conversely, another recent review studied the sensorimotor deficits associated with FAI and concluded that peroneal reaction time was not affected

in FAI ankles compared with healthy ankles (71). However, the authors associated FAI with deficits in postural control and ankle joint position sense.

Muscle activation imbalances between medial and lateral sides of the quadriceps and hamstrings. Imbalances between the medial and lateral parts of a muscle have been described as risk factors for injury, especially in both the quadriceps (27,83) and hamstrings (106). Investigations into the activation pattern of the quadriceps in a high-risk movement pattern known to result in ACL injury in both sexes indicated that females activate a higher proportion of the lateral side of the quadriceps muscle compared with males, which may contribute to dynamic knee valgus and facilitate ACL tear (83). Moreover, this injury mechanism is compounded by predominance in activation of the lateral part of the hamstring muscles (106). With regard to patellofemoral pain syndrome, early evidence indicates that a lower magnitude and longer activation of the vastus medialis as compared with the vastus lateralis is associated with a higher incidence of this condition (20,27).

Diminished muscle coactivation between agonist and antagonist muscles. Compared with agonistic

Table 1 Neuromuscular injury risk factors and related neuromuscular training component in youth active population	
Potential neuromuscular risk factors in youth	Neuromuscular training focus
Muscle fatigue	Strength and power muscle endurance
	High-intensity interval training
	Progression to training in fatigue conditions
Altered timing and magnitude of muscle activation	Strength/power (plyometric)
Delay in reaction time of peroneus muscles	Dynamic stabilization
Muscle activation imbalances between medial and lateral sides of the quadriceps and hamstrings	Proprioception
Diminished muscle coactivation between agonist and antagonist muscles	Awareness muscle activation exercises
Decreased hip muscle activation	
Deficits in trunk stability and muscle activation	
Strength weakness	Agonist/antagonist strength balance
	Compensatory muscles strength
	Strength/power training
	Overload eccentric training
Frontal plane knee control: dynamic valgus	Good technique (low limb alignment with trunk-hip-knee-ankle flexion) during landings, change of directions, and deceleration actions
Neuromuscular imbalances between limbs	Symmetry technique in bilateral tasks
	Unilateral tasks
Inadequate muscle stiffness	Strength
	Power
Deficits in postural stability	Balance → dynamic stabilization
Altered proprioception	Fundamental movement skills
	Balance/dynamic stabilization
	Functional mobility
Feed-forward mechanism	Training with unexpected actions
	Task variability
	Agility open tasks
	Coordination abilities

activation in isolation, the coactivation of agonist and antagonist muscles has been associated with joint stabilization and the reduction of ligament loading due to increased joint stiffness (60). Diminished coactivation between quadriceps and hamstrings has been related to risk factors associated with ACL injuries (32,47). Moreover, the coactivation of these 2 muscles during cutting, landing, or deceleration not only provides knee joint stability but also protects from excessive anterior tibial displacement and dynamic knee valgus (32,47,60). However, current literature also emphasizes the importance of coactivation for both the transverse abdominis and multifidus muscles during functional tasks for the prevention and treatment of low-back pain (63).

Decreased hip muscle activation. Decreased hip muscle activation has been associated with risk factors related to ACL rupture (127) and concurrent patellofemoral pain syndrome (123). Research suggests that the decreased hip muscle activity and increased quadriceps activity exhibited by females compared with males during single-leg landings may be an important contributor to the increased susceptibility of female athletes to noncontact ACL injuries (45,127). Despite limited data about the relationship between hip muscle activation in fatigue conditions, Patrek (97) observed that gluteus medius activation did not decrease when fatigue occurred during single-leg landing. However, the activation of this muscle delayed after the protocol. Related to this last point, and from an injury standpoint, the timing of activation is more relevant than the magnitude of the activation (22). Conversely, females with patellofemoral pain have demonstrated delayed and shorter gluteus medius activation compared with females without knee pain during running (123).

Deficits in trunk stability and muscle activation. In the past, deficits of the

sensorimotor system in the body's core (spine, hips, and pelvis) have been associated with an increase in lower limb injury (73,126). Alterations in the neuromuscular control of the trunk have been associated with increased knee valgus, and this puts the knee joint at risk for injury because of transmission of higher loads (73). Similarly, Zazulak et al. observed that decreased proprioception and neuromuscular control of the trunk could predict an increased risk of knee injuries in females, but not in males. Moreover, this sex difference is related to the aforementioned higher deficits in neuromuscular control described in females when compared with males (126). It has also been suggested that deficits in the activation of hip and trunk muscles associated with pubertal age may increase injury risk (73). As previously mentioned, the pubertal age group is especially vulnerable to neuromuscular deficits (86).

STRENGTH DEFICITS

Dynamic joint stability depends on passive (ligaments and articular geometry) and active (muscles and neuromuscular control) restraints of the joint. When active restraints are compromised, there may be an increased risk of injury (111). A lack of muscular strength may lead to poor neuromuscular control, which is a precursor to injury. Research indicates that strength deficits may be related to both acute (e.g., ACL rupture or hamstrings strains) and chronic (e.g., patellofemoral pain) lower limb sports injuries (43–58).

Previous research indicates that strength deficits are related to both antagonist/agonist and contralateral strength imbalances. Despite the abundance of literature on strength imbalances around the knee, evidence regarding their contribution to ACL and hamstring injuries remains inconclusive (2,44,92). Research indicates that a hamstring-to-quadriceps ratio lower than 60% can predispose an athlete to ACL injury (44,80). Lower levels of

hamstring strength compared with quadriceps strength during landing and cutting maneuvers increases anterior shear stress on the tibia, which increases the load on the ACL. Another major risk factor for ACL injury is strength asymmetries between limbs (47,88,94). Research shows that bilateral differences of 15% or more in concentric and/or eccentric actions of the hamstrings increases the risk of injury to these muscles (21). Similarly, a concentric hamstrings/quadriceps strength ratio (on at least one leg) of less than 0.47–0.45 and a mixed ratio of eccentric hamstring/concentric quadriceps strength ratio lower than 0.80–0.89 (measured by an isokinetic device) also increase the risk of hamstring injury in soccer players (21). However, it is important to mention that, in addition to imbalances, both absolute and relative strength should be considered (122). An appropriate hamstring/quadriceps ratio is irrelevant if the athlete displays inferior strength. Conversely, gaining absolute strength without the development of antagonistic muscle action may further propagate imbalances that increase inherent risk factors (80).

In addition to such deficits, it is important to note that there is a possibility that decreased eccentric muscle strength may lead to injury. This assertion is supported by research that connects eccentric strength deficits in the quadriceps muscle to tendinopathies in athletes (102). The forces necessary to resist knee extension and start hip extension during maximal sprints could surpass the tolerance of the muscle-tendon unit when these ratios of strength are not achieved because of hamstring strength deficits (3). The previously reported threshold for the critical difference between legs is 10–15% measured using isokinetic strength and vertical jump data (50,62,72,88,95).

Although knee strength deficits are critical, it is also important to note the relationship between hip muscle strength deficits and sports injury. Deficits in strength combined with hip

abduction, hip external rotation, and hip extension are risk factors associated with patellofemoral pain syndrome (97), ACL injury (53), and iliotibial band syndrome (38). Furthermore, the achievement of an eccentric hip adduction/abduction strength ratio of more than 90% has been suggested to play an important role in the treatment and prevention of groin injuries in soccer players (117).

Finally, deficits in power have also been associated with an increased rate of injury (124). Low levels of power may underlie increased electromechanical delay, and this causes a decrease in muscle reaction speed in the context of sports actions.

FRONTAL PLANE KNEE CONTROL: DYNAMIC VALGUS

Altered knee joint biomechanics, especially those that lead to lack of control strategies in the frontal plane of the knee during landing, cutting, and deceleration, are considered major risk factors for knee injuries, including ACL rupture (47,95) and patellofemoral pain (75,79). The implications of biomechanical abnormalities in the knee differ depending on sex. Female athletes exhibit a 4-fold to 6-fold increase in incidence of ACL injuries over male athletes (46). Other research on male and female youth athletes (11–19 years old) has demonstrated a predominant dynamic knee valgus during actions at high risk for injury, including landing (5,87,108). However, it is important to isolate the relationship between dynamic valgus and age. In the prepubescent population, the number of injuries may be lower because younger athletes have less body mass, shorter joint lever arms and do not generate as much power, and, therefore, do not exhibit as much dynamic valgus load as their more mature counterparts. As a result, sport actions are developed at a lower power in younger athletes than in older athletes. This may account for the lower rate of injury in younger athletes (10,120) and reaffirms the notion that INT programs should be initiated during

prepubescence to establish proper technique before athletes begin to generate enough power to increase their risk of injury (85).

NEUROMUSCULAR IMBALANCES BETWEEN LIMBS

Neuromuscular asymmetry of the lower limbs is associated with injury and can be used as a predictive tool to detect athletes who may be at risk for injury or reinjury (94,107). Although lower limb asymmetries are normal in healthy athletes, research suggests that a difference in strength and power greater than 10–15% between legs is the threshold to consider for increased risk of injury (50,62,72,88,95). Similarly, it has been shown that differences between limbs of <15% in the single-leg vertical jump should be considered as the physiological norm in youth (16,35). It is important to note that lower limb asymmetries in strength, coordination, and postural control are more common in female athletes than males athletes, especially during adolescence (30,34,76). Testing procedures to identify and track neuromuscular asymmetries should be implemented to detect at-risk athletes, and INT programs can help guide corrective strategies to address asymmetries and potentially reduce the risk of injury.

INADEQUATE MUSCLE STIFFNESS

Muscle stiffness, defined as a muscle's ability to resist lengthening and contrary to compliance (26), is essential for both the maintenance of joint stability (40) and the ability to generate power (14). To better understand this concept, it is important to differentiate between active and passive stiffness (40). At low applied loads, the passive structures of the joint (e.g., ligaments, capsules) provide enough stability. However, during sport activities, joint forces go beyond the stabilizing capacity of the joint capsule and ligaments and muscles are recruited to stabilize the joint (39). Active muscle stiffness is considered an essential component of joint stability during functional and sporting activities, preventing against

musculoskeletal injuries (111). Moreover, active muscle stiffness is related to active joint stiffness, which can be voluntarily controlled through muscle recruitment (40).

The relationship between active stiffness and injury is still unclear because of a lack of research (14). There is some evidence to suggest that there is an “optimal” amount of stiffness. Low levels of stiffness have been associated with soft tissue injuries (40). However, too much stiffness may be related to high levels of peak forces and loading rates, which can be associated to bony injuries (14). In terms of performance, an athlete with greater muscle stiffness in the lower extremity more efficiently uses and reuses stored elastic energy during stretch-shortening cycle exercises (jumping, running, and hopping) (13).

Recent evidence demonstrates that females have reduced muscle stiffness compared with males of the same age (125). Power and strength training are needed to improve motor unit recruitment and, consequently, to improve muscle stiffness (69). To further improve muscular stiffness, neuromuscular training for youth should emphasize explosive power and eccentric loading.

In addition, muscle stiffness depends on maturation/developmental status. As previously mentioned, puberty is accompanied by hormonal changes, which lead to an increase in muscle stiffness, especially in men (98). In this case, concomitant training methods to improve muscle stiffness and compliance should be applied (13).

DEFICITS IN POSTURAL STABILITY

Postural stability, also known as dynamic balance, is dependent on the integration of perceptual information and neuromuscular control strategies to regulate the stability of the body's center of mass during dynamic tasks (103,104). Currently, the relationship between balance and risk of lower extremity injury is unclear. Although some research indicates that there is no relationship between the 2

(7,66), most research has found significant correlations between deficits in postural stability and injury incidence (64,99,118), especially in individuals with history of injury (66,95). The high variability of methods used to assess postural stability could partially explain the contrasting results in existing research.

In youth athletes, an individual's inability to control and maintain postural stability during static and dynamic tasks has been associated with risk of injury and reinjury, especially in the ankle (64,99) and knee (95,99). Plisky (99) showed that high school players with a greater balance capacity (a superior score in anterior and lateral distances performing the star excursion balance test) were 2.5 times less likely to sustain a lower limb injury. This finding demonstrates the importance of incorporating balance training into strength and conditioning programs in youth athletes.

ALTERED PROPRIOCEPTION

The proprioceptive system is a part of the sensorimotor system that incorporates afferent and efferent signals and serves as a conduit for the integration of sensorimotor control during the maintenance of functional joint stability (103). Although visual and vestibular input contributes to the performance of athletic actions, the peripheral mechanoreceptors are typically influenced most by injury and are thus the most modifiable from a training perspective (49). There is currently no consensus on the exact definition of proprioception in the literature (49,103,105). For the purposes of this review, we will define proprioception as the type of sensorimotor sensitivity that contributes to the maintenance of dynamic joint stability, achieved by the detection of pressure, tension, and length variations of the different muscles and articular tissues (37).

Diminished proprioception has been associated with abnormal biomechanics, which, in turn, predispose

athletes to injury. Diminished proprioception has been shown to be a risk factor for many injuries, including ACL ruptures (46,126), patellofemoral pain syndrome (4), and chronic ankle instability (65). In addition, impaired proprioception in previously injured joints predisposes athletes to reinjury (95,99). To ensure the establishment of proper technique and to decrease the risk of injury in youth athletes, proprioception training should be the first step in the implementation of INT programs. A strong base in proprioception will provide the foundation to progress athletes to more challenging, intense, and sport-specific actions. Proprioception training will be a prominent point of discussion in part II of this commentary.

FEED-FORWARD MECHANISMS

The dynamic stabilization of the joints during sport activities depends on both feedback (reflex response) and feed-forward (preactivation or anticipation) neuromuscular control mechanisms (8,9,103,104). Feedback control is characterized by a continual processing of afferent information, which provides a reflexive muscle response on a moment-to-moment basis (103). During sport activities, especially in team or situational sports where the context is continuously changing, feedback control plays a limited role in injury prevention. Although less understood as a process for injury prevention in fast-paced situations, feedback control is necessary for maintaining posture and protecting joints during slower, more predictable actions (37).

Conversely, feed-forward (i.e., prospective) control has been described as anticipatory actions taken before sensory detection of a homeostatic disruption (103). That is, muscle preactivation has the ability to protect joint structures from a potentially injurious load. This neuromuscular control mechanism depends on previous movement experiences, previous exposures to external stressors, and learned perceptual-motor relationships (28).

The continuous information provided via the feed-forward mechanism facilitates the motor learning process and provides an opportunity to make adjustments to avoid injury.

Although feedback control is related to mechanoreceptor detection of altered support surface, feed-forward control is related to anticipatory actions from previous experience and learned relationships between the athlete and the environment (28,103). This indicates that the feed-forward control allows for the necessary activation of the muscles stabilizing joints during the preparatory phase of movements before potential injury. For this reason, feed-forward is considered as the most important factor in maintaining dynamic stabilization during landing, deceleration, and cutting movements (9,51,103).

Current research emphasizes the importance of unanticipated movements that function to prompt the maintenance of appropriate biomechanics during the execution of high-risk tasks that may cause injury, including landing and cutting maneuvers (8,51,57). Cutting maneuvers performed without adequate planning (unanticipated actions) increase risk factors for noncontact knee ligament injury compared with predetermined actions (7,9). These risk factors include increased external varus/valgus and internal/external rotation torques applied to the knee. In addition, Ford et al. (31) showed greater altered kinematics in female adolescent athletes compared with similar male athletes and suggested that differences in kinematics may contribute to the greater rate of ACL injury documented in female athletes. Other research on unanticipated actions has shown that the inclusion of an opponent during cutting and landing actions increased the mechanical load in the knee (57,68). Furthermore, research indicates that feedback control of peroneus muscles is not affected in the presence of chronic ankle instability, indicating that the burden of negative effects falls mainly on feed-forward

control (51). Consequently, when designing INT programs, it is essential to progress to unanticipated and opposed tasks to train the feed-forward system, especially in youth athletes because of their high neural plasticity (11).

CONCLUSIONS

This review of the risk factors provides important background information that is needed to design neuromuscular tasks that can be adapted to each athlete and sport, which will be useful to coaches, strength and conditioning professionals, and physiotherapists.

Research indicates that neuromuscular risk factors can be modified through INT programs. Although the optimal training methods adapted from each specific population are untested, evidence exists to support the implementation of interventions to prevent, treat, or return youth athletes to sport after certain lower limb injuries. When implementing an INT program for youth athletes, it is important to progress exercises in intensity and difficulty while maintaining an emphasis on the correct lower limb biomechanics to minimize joint loads during sport-specific actions. During INT development, it is important for individuals creating INT programs to select appropriate tasks for each athlete and to provide constant feedback about how the athlete is executing each exercise. Finally, the current evidence indicates that prescreening with valid and reliable tests can help identify neuromuscular risk factors that will benefit from targeted training and can help optimize the efficiency and effectiveness of INT programs in youth.

Conflicts of Interest and Source of Funding: We acknowledge funding support from National Institutes of Health/NLAMS Grant R21AR065068-01A1 and U01AR067997.



Azahara Fort-Vanmeerhaeghe

is an Associate Professor at the School of Health and Sport Sciences (EUSES, University of Girona, Spain) and a strength and

conditioning coach of elite female youth basketball players (Segle XXI team, Catalan Federation of Basketball).



Daniel Romero-Rodriguez

is an Associate Professor at the School of Health and Sport Sciences (EUSES, University of Girona, Spain) and the Sporting and

Technical Director of ReSport Clinic (Barcelona, Spain) and Novaelite Sport Center (Barcelona, Spain).



Alicia M. Montalvo

is an Assistant Professor of Athletic Training at Florida International University.



Adam W. Kiefer

is an Assistant Professor in the Division of Sports Medicine at Cincinnati Children's Hospital Medical Center and holds appointments in the

Department of Pediatrics in the College of Medicine at the University of Cincinnati, and also at the Center for

Cognition, Action and Perception in the Department of Psychology at the University of Cincinnati.



Rhodri S. Lloyd

is a Senior Lecturer in Strength and Conditioning at Cardiff Metropolitan University.



Gregory D. Myer

is Director of Research and the Human Performance Laboratory for the Division of Sports Medicine at Cincinnati Children's Hos-

pital Medical Center and holds primary academic appointments in the Departments of Pediatrics and Orthopaedic Surgery within the College of Medicine at University of Cincinnati.

REFERENCES

- Adirim TA and Cheng TL. Overview of injuries in the young athlete. *Sports Med* 33: 75–81, 2003.
- Arnason A, Andersen TE, Holme I, Engebretsen L, and Bahr R. Prevention of hamstring strains in elite soccer: An intervention study. *Scand J Med Sci Sports* 18: 40–48.
- Bahr R and Holme I. Risk factors for sports injuries—a methodological approach. *Br J Sports Med* 37: 384–392, 2003.
- Baker V, Bennell K, Stillman B, Cowan S, and Crossley K. Abnormal knee joint position sense in individuals with patellofemoral pain syndrome. *J Orthop Res* 20: 208–214, 2002.
- Barber-Westin SD, Noyes FR, and Galloway M. Jump-land characteristics and muscle strength development in young athletes: A gender comparison of 1140 athletes 9 to 17 years of age. *Am J Sports Med* 34: 375–384, 2006.
- Bergeron MF, Mountjoy M, Armstrong N, Chia M, Cote J, and Emery CA, et al.

International Olympic Committee consensus statement on youth athletic development. *Br J Sports Med* 49: 843–851, 2015.

- Bernier JN, Perrin DH, and Rijke A. Effect of unilateral functional instability of the ankle on postural sway and inversion and eversion strength. *J Athl Train* 32: 226–232, 1997.
- Besier T and Lloyd D. Anticipatory effects on knee joint loading during running and cutting maneuvers. *Med Sci Sports Exerc* 1: 1176–1181, 2001.
- Besier TF, Lloyd DG, and Ackland TR. Muscle activation strategies at the knee during running and cutting maneuvers. *Med Sci Sports Exerc* 35: 119–127, 2003.
- Bloemers F, Collard D, Paw MCA, Van Mechelen W, Twisk J, and Verhagen E. Physical inactivity is a risk factor for physical activity-related injuries in children. *Br J Sports Med* 46: 669–674, 2012.
- Borotikar BS, Newcomer R, Koppes R, and McLean SG. Combined effects of fatigue and decision making on female lower limb landing postures: Central and peripheral contributions to ACL injury risk. *Clin Biomech* 23: 81–92, 2008.
- Brazen DM, Todd MK, Ambegaonkar JP, Wunderlich R, and Peterson C. The effect of fatigue on landing biomechanics in single-leg drop landings. *Clin J Sport Med* 20: 286–292, 2010.
- Brazier J, Bishop C, Chris S, Antrobus M, Read P, and Turner A. Lower extremity stiffness: Effects on performance and injury and implications for training. *Strength Cond J* 36: 103–112, 2014.
- Butler RJ, Crowell HP, and Davis IM. Lower extremity stiffness: Implications for performance and injury. *Clin Biomech* 18: 511–517, 2003.
- Caine D, Purcell L, and Maffulli N. The child and adolescent athlete: A review of three potentially serious injuries. *Sports Sci Med Rehabil* 6: 22, 2014.
- Ceroni D, Martin XE, Delhumeau C, and Farpour-Lambert NJ. Bilateral and gender differences during single-legged vertical jump performance in healthy teenagers. *J Strength Cond Res* 26: 452–457, 2012.
- Chappell JD, Herman DC, Knight BS, Kirkendall DT, Garrett WE, and Yu B. Effect of fatigue on knee kinetics and kinematics in stop-jump tasks. *Am J Sports Med* 33: 1022–1029, 2005.

18. Chappell JD and Limpisvasti O. Effect of a neuromuscular training program on the kinetics and kinematics of jumping tasks. *Am J Sports Med* 36: 1081–1086, 2008.
19. Collard DCM, Verhagen EALM, van Mechelen W, Heymans MW, and Chinapaw MJM. Economic burden of physical activity-related injuries in Dutch children aged 10-12. *Br J Sports Med* 45: 1058–1063, 2011.
20. Cowan SM, Bennell KL, Hodges PW, Crossley KM, and McConnell J. Delayed onset of electromyographic activity of vastus medialis obliquus relative to vastus lateralis in subjects with patellofemoral pain syndrome. *Arch Phys Med Rehabil* 82: 183–189, 2001.
21. Croisier JL, Ganteaume S, Binet J, Genty M, and Ferret JM. Strength imbalances and prevention of hamstring injury in professional soccer players: A prospective study. *Am J Sports Med* 36: 1469–1475, 2008.
22. De Ste Croix MBA, Priestley AM, Lloyd RS, and Oliver JL. ACL injury risk in elite female youth soccer: Changes in neuromuscular control of the knee following soccer-specific fatigue. *Scand J Med Sci Sports*, 2014. Epub ahead of print.
23. Dipla K, Tsirini T, Zafeiridis A, Manou V, Dalamitros A, and Kellis E, et al. Fatigue resistance during high-intensity intermittent exercise from childhood to adulthood in males and females. *Eur J Appl Physiol* 106: 645–653, 2009.
24. Emery CA, Hagel B, and Morrongiello BA. Injury prevention in child and adolescent sport: Whose responsibility is it? *Clin J Sport Med* 16: 514–521, 2006.
25. Emery CA, Roy TO, Whittaker JL, Nettel-Aguirre A, and van Mechelen W. Neuromuscular training injury prevention strategies in youth sport: A systematic review and meta-analysis. *Br J Sports Med* 49: 865–870, 2015.
26. Enoka RM and Duchateau J. Muscle fatigue: What, why and how it influences muscle function. *J Physiol* 586: 11–23, 2008.
27. Fagan V and Delahunt E. Patellofemoral pain syndrome: A review on the associated neuromuscular deficits and current treatment options. *Br J Sports Med* 42: 789–795, 2008.
28. Fajen BR, Riley MA, and Turvey MT. Information, affordances, and the control of action in sport. *Int J Sport Psychol* 40: 79–107, 2009.
29. Flynn JM, Lou JE, and Ganley TJ. Prevention of sports injuries in children. *Curr Opin Pediatr* 14: 719–722, 2012.
30. Ford KR, Myer GD, and Hewett TE. Valgus knee motion during landing in high school female and male basketball players. *Med Sci Sports Exerc* 35: 1745–1750, 2003.
31. Ford KR, Myer GD, Toms HE, and Hewett TE. Gender differences in the kinematics of unanticipated cutting in young athletes. *Med Sci Sports Exerc* 37: 124–129, 2005.
32. Ford KR, van den Bogert J, Myer GD, Shapiro R, and Hewett TE. The effects of age and skill level on knee musculature co-contraction during functional activities: A systematic review. *Br J Sports Med* 42: 561–566, 2008.
33. Forestier N, Teasdale N, and Nougier V. Alteration of the position sense at the ankle induced by muscular fatigue in humans. *Med Sci Sports Exerc* 34: 117–122, 2002.
34. Fort-Vanmeerhaeghe A, Gual G, Romero-Rodriguez D, Costa L, and Walsh M. Explosive strength asymmetry of the lower limb in volleyball and basketball players. Presented at: 18th annual Congress of the European College of Sport; June 16–18, 2013; Barcelona, Spain.
35. Fort-Vanmeerhaeghe A, Montalvo AM, Sitjà-Rabert M, Kiefer AW, and Myer GD. Neuromuscular asymmetries in the lower limbs of elite female youth basketball players and the application of the skillful limb model of comparison. *Phys Ther Sport* 16: 317–323, 2015.
36. Fort-Vanmeerhaeghe A and Romero-Rodriguez D. Análisis de los factores de riesgo neuromusculares de las lesiones deportivas. *Apunt Med Sport* 48: 109–120, 2013.
37. Fort-Vanmeerhaeghe A and Romero-Rodriguez D. Rol del sistema sensoriomotor en la estabilidad articular durante las actividades deportivas. *Apunt Med Sport* 48: 69–76, 2012.
38. Fredericson M, Cookingham CL, Chaudhari aM, Dowdell BC, Oestreich N, and Sahrmann SA. Hip abductor weakness in distance runners with iliotibial band syndrome. *Clin J Sport Med* 10: 169–175, 2000.
39. Granata KP, Padua DA, and Wilson SE. Gender differences in active musculoskeletal stiffness. Part II. Quantification of leg stiffness during functional hopping tasks. *J Electromyogr Kinesiol* 12: 127–135, 2002.
40. Granata KP, Wilson SE, and Padua DA. Gender differences in active musculoskeletal stiffness. Part I. Quantification in controlled measurements of knee joint dynamics. *J Electromyogr Kinesiol* 12: 119–126, 2002.
41. Herman K, Barton C, Malliaras P, and Morrissey D. The effectiveness of neuromuscular warm-up strategies, that require no additional equipment, for preventing lower limb injuries during sports participation: A systematic review. *BMC Med* 10: 75, 2012.
42. Hewett TE, Ford KR, Hoogenboom BJ, and Myer GD. Understanding and preventing acl injuries: Current biomechanical and epidemiologic considerations—Update 2010. *N Am J Sports Phys Ther* 5: 234–251, 2010.
43. Hewett TE, Ford KR, and Myer GD. Anterior cruciate ligament injuries in female athletes: Part 2, a meta-analysis of neuromuscular interventions aimed at injury prevention. *Am J Sports Med* 34: 490–498, 2006.
44. Hewett TE, Lindenfeld TN, Riccobene JV, and Noyes FR. The effect of neuromuscular training on the incidence of knee injury in female athletes. A prospective study. *Am J Sports Med* 27: 699–706, 1999.
45. Hewett TE and Myer GD. The mechanistic connection between the trunk, hip, knee, and anterior cruciate ligament injury. *Exerc Sport Sci Rev* 39: 161–166, 2011.
46. Hewett TE, Myer GD, and Ford KR. Anterior cruciate ligament injuries in female athletes: Part 1, mechanisms and risk factors. *Am J Sports Med* 2006;34: 299–311.
47. Hewett TE, Myer GD, Ford KR, Heidt RS, Colosimo AJ, and McLean SG, et al. Biomechanical measures of neuromuscular control and valgus loading of the knee predict anterior cruciate ligament injury risk in female athletes: A prospective study. *Am J Sports Med* 33: 492–501, 2005.
48. Hewett TE, Myer GD, Kiefer AW, and Ford KR. Longitudinal increases in knee abduction moments in females during adolescent growth. *Med Sci Sports Exerc* 47: 2579–2585, 2015.
49. Hewett TE, Paterno MV, and Myer GD. Strategies for enhancing proprioception and neuromuscular control of the knee. *Clin Orthop Relat Res* 402: 76–94, 2002.

50. Hewit J, Cronin J, and Hume P. Asymmetry in multi-directional jumping tasks. *Phys Ther Sport* 13: 238–242, 2012.
51. Holmes A and Delahunt E. Treatment of common deficits associated with chronic ankle instability. *Sport Med* 39: 207–224, 2009.
52. Hübscher M, Zech A, Pfeifer K, Hänsel F, Vogt L, and Banzer W. Neuromuscular training for sports injury prevention: A systematic review. *Med Sci Sports Exerc* 42: 413–421, 2010.
53. Imwalle L and Myer G. Relationship between hip and knee kinematics in athletic women during cutting maneuvers: A possible link to noncontact anterior cruciate ligament injury and. *J Strength Cond Res* 23: 2223–2230, 2009.
54. Ireland ML, Willson JD, Ballantyne BT, and Davis IM. Hip strength in females with and without patellofemoral pain. *J Orthop Sports Phys Ther* 33: 671–676, 2003.
55. Kernozek TW, Torry MR, and Iwasaki M. Gender differences in lower extremity landing mechanics caused by neuromuscular fatigue. *Am J Sports Med* 36: 554–565, 2008.
56. Kraemer W and Fleck S. Strength training for young athletes. Kinetics H, ed. Champaign, IL, 2005.
57. Kristianslund E and Krosshaug T. Comparison of drop jumps and sport-specific sidestep cutting: Implications for anterior cruciate ligament injury risk screening. *Am J Sports Med* 41: 684–688, 2013.
58. Lankhorst NE, Bierma-Zeinstra SMA, and van Middelkoop M. Risk factors for patellofemoral pain syndrome: A systematic review. *J Orthop Sports Phys Ther* 42: 81–94, 2012.
59. Leek D, Carlson JA, Cain KL, Henrichon S, Rosenberg D, and Patrick K, et al. Physical activity during youth sports practices. *Arch Pediatr Adolesc Med* 165: 294–299, 2011.
60. Lloyd D, Buchanan T, and Besier T. Neuromuscular biomechanical modeling to understand knee ligament loading. *Med Sci Sports Exerc* 37: 1939–1947, 2005.
61. Löfvenberg R, Kärrholm J, Sundelin G, and Ahlgren O. Prolonged reaction time in patients with chronic lateral instability of the ankle. *Am J Sports Med* 23: 414–417, 1995.
62. McElveen MT, Riemann BL, and Davies GJ. Bilateral comparison of propulsion mechanics during single-leg vertical jumping. *J Strength Cond Res* 24: 375–381, 2010.
63. McGill S. Core training: Evidence translating to better performance and injury prevention. *Strength Cond J* 32: 33–46, 2010.
64. McGuine TA, Greene JJ, Best T, and Levenson G. Balance as a predictor of ankle injuries in high school basketball players. *Clin J Sport Med* 10: 239–244, 2000.
65. McKeon PO, Booi MJ, Branam B, Johnson DL, and Mattacola CG. Lateral ankle ligament anesthesia significantly alters single limb postural control. *Gait Posture* 32: 374–377, 2010.
66. McKeon PO and Hertel J. Systematic review of postural control and lateral ankle instability, part I: Can deficits be detected with instrumented testing. *J Athl Train* 43: 293–304, 2008.
67. McLean SG, Fellin RE, Felin RE, Suedekum N, Calabrese G, and Passerallo A, et al. Impact of fatigue on gender-based high-risk landing strategies. *Med Sci Sports Exerc* 39: 502–514, 2007.
68. McLean SG, Lipfert SW, and van den Bogert AJ. Effect of gender and defensive opponent on the biomechanics of sidestep cutting. *Med Sci Sports Exerc* 36: 1008–1016, 2004.
69. McMahon J, Comfort P, and Pearson S. Lower limb stiffness: Considerations for female athletes. *Strength Cond J* 34: 70–73, 2012.
70. Menacho Mde O, Pereira HM, Oliveira BIR, Chagas LMPM, Toyohara MT, and Cardoso JR. The peroneus reaction time during sudden inversion test: Systematic review. *J Electromyogr Kinesiol* 20: 559–565, 2010.
71. Munn J, Sullivan SJ, and Schneiders AG. Evidence of sensorimotor deficits in functional ankle instability: A systematic review with meta-analysis. *J Sci Med Sport* 13: 2–12, 2010.
72. Munro AG and Herrington LC. Between-session reliability of four hop tests and the agility T-test. *J Strength Cond Res* 25: 1470–1477, 2011.
73. Myer G, Chu D, Brent J, and Hewett T. Trunk and hip control neuromuscular training for the prevention of knee joint injury. *Clin Sports Med* 27: 1–23, 2008.
74. Myer G and Faigenbaum A. Pediatric physical activity exercise is sports medicine in youth : Integrative neuromuscular training to optimize motor development. *KRONOS X*: 39–48, 2011.
75. Myer G, Ford K, Di Stasi S, Barber K, Micheli L, and Hewett T. High knee abduction moments are common risk factors for patellofemoral pain (PFP) and anterior cruciate ligament (ACL) injury in girls: Is PFP itself a predictor for subsequent ACL injury? *Br J Sports Med* 49: 118–122, 2015.
76. Myer GD, Brent JL, Ford KR, and Hewett TE. Real-time assessment and neuromuscular training feedback techniques to prevent ACL injury in female athletes. *Strength Cond J* 33: 21–35, 2011.
77. Myer GD, Faigenbaum AD, Chu DA, Falkel J, Ford KR, and Best TM, et al. Integrative training for children and adolescents: Techniques and practices for reducing sports-related injuries and enhancing athletic performance. *Phys Sportsmed* 39: 74–84, 2011.
78. Myer GD, Faigenbaum AD, Ford KR, Best TM, Bergeron MF, and Hewett TE. When to initiate integrative neuromuscular training to reduce sports-related injuries and enhance health in youth? *Curr Sports Med Rep* 10: 157–166, 2011.
79. Myer GD, Ford KR, Barber Foss KD, Goodman A, Ceasar A, and Rauh MJ, et al. The incidence and potential pathomechanics of patellofemoral pain in female athletes. *Clin Biomech* 25: 700–707, 2010.
80. Myer GD, Ford KR, Barber Foss KD, Liu C, Nick TG, and Hewett TE. The relationship of hamstrings and quadriceps strength to anterior cruciate ligament injury in female athletes. *Clin J Sport Med* 19: 3–8, 2009.
81. Myer GD, Ford KR, Brent JL, and Hewett TE. Differential neuromuscular training effects on ACL injury risk factors in “high-risk” versus “low-risk” athletes. *BMC Musculoskelet Disord* 8: 39, 2007.
82. Myer GD, Ford KR, Brent JL, and Hewett TE. The effects of plyometric vs. dynamic stabilization and balance training on power, balance, and landing force in female athletes. *J Strength Cond Res* 20: 345–353, 2006.
83. Myer GD, Ford KR, and Hewett TE. The effects of gender on quadriceps muscle activation strategies during a maneuver that mimics a high ACL injury risk position. *J Electromyogr Kinesiol* 15: 181–189, 2005.

84. Myer GD, Ford KR, Palumbo JP, and Hewett TE. Neuromuscular training improves performance and lower-extremity biomechanics in female athletes. *J Strength Cond Res* 19: 51–60, 2005.
85. Myer GD, Lloyd RS, and Brent JLFA. How young is too young to start training? *ACSMs Heal Fit J* 17: 14–23, 2013.
86. Myer GD, Sugimoto D, Thomas S, and Hewett TE. The influence of age on the effectiveness of neuromuscular training to reduce anterior cruciate ligament injury in female athletes: A meta-analysis. *Am J Sports Med* 41: 203–215, 2013.
87. Noyes F, Barber-Westin S, Fleckenstein C, Walsh C, and West J. The drop-jump screening test: Difference in lower limb control by gender and effect of neuromuscular training in female athletes. *Am J Sports Med* 33: 197–207, 2005.
88. Noyes FR, Barber SD, and Mangine RE. Abnormal lower limb symmetry determined by function hop tests after anterior cruciate ligament rupture. *Am J Sports Med* 19: 513–518, 1991.
89. Noyes FR, Barber-Westin SD, Smith ST, Campbell T, and Garrison TT. A training program to improve neuromuscular and performance indices in female high school basketball players. *J Strength Cond Res* 26: 709–719, 2012.
90. Noyes FR, Barber-Westin SD, Tutalo Smith ST, and Campbell T. A training program to improve neuromuscular and performance indices in female high school soccer players. *J Strength Cond Res* 27: 340–351, 2013.
91. Olsen OE, Myklebust G, Engebretsen L, Holme I, and Bahr R. Exercises to prevent lower limb injuries in youth sports: Cluster randomised controlled trial. *BMJ* 330: 449, 2005.
92. Orchard J, Marsden J, Lord S, and Garlick D. Preseason hamstring muscle weakness associated with hamstring muscle injury in Australian footballers. *Am J Sports Med* 25: 81–85, 1997.
93. Pate RR and O'Neill JR. Youth sports programs: Contribution to physical activity. *Arch Pediatr Adolesc Med* 165: 369–370, 2011.
94. Paterno MV, Ford KR, Myer GD, Heyl R, and Hewett TE. Limb asymmetries in landing and jumping 2 years following anterior cruciate ligament reconstruction. *Clin J Sport Med* 17: 258–262, 2007.
95. Paterno MV, Schmitt LC, Ford KR, Rauh MJ, Myer GD, and Huang B, et al. Biomechanical measures during landing and postural stability predict second anterior cruciate ligament injury after anterior cruciate ligament reconstruction and return to sport. *Am J Sports Med* 38: 1968–1978, 2010.
96. Paterno MV, Taylor-Haas JA, Myer GD, and Hewett TE. Prevention of overuse sports injuries in the young athlete. *Orthop Clin North Am* 44: 553–564, 2013.
97. Patrek MF, Kernozek TW, Willson JD, Wright GA, and Doberstein ST. Hip-abductor fatigue and single-leg landing mechanics in women athletes. *J Athl Train* 46: 31–42, 2011.
98. Philippaerts RM, Vaeyens R, Janssens M, Van Renterghem B, Matthyss D, and Craen R, et al. The relationship between peak height velocity and physical performance in youth soccer players. *J Sports Sci* 24: 221–230, 2006.
99. Plisky P. Star excursion balance test as a predictor of lower extremity injury in high school basketball players. *J Orthop Sports Phys Ther* 36: 911–919, 2006.
100. Prins MR and van der Wurff P. Preseason hamstring muscle weakness associated with hamstring muscle injury in Australian footballers. *Aust J Physiother* 55: 9–15, 2009.
101. Ratel S, Duché P, and Williams Ca. Muscle fatigue during high-intensity exercise in children. *Sports Med* 36: 1031–1065, 2006.
102. Reinking M. Tendinopathy in athletes. *Phys Ther Sport* 13: 3–10, 2012.
103. Riemann BL and Lephart SM. The sensorimotor system, part I: The physiologic basis of functional joint stability. *J Athl Train* 37: 71–79, 2002.
104. Riemann BL and Lephart SM. The sensorimotor system, Part II: The role of proprioception in motor control and functional joint stability. *J Athl Train* 37: 80–84, 2002.
105. Roberts D, Ageberg E, Andersson G, and Fridén T. Clinical measurements of proprioception, muscle strength and laxity in relation to function in the ACL-injured knee. Knee surgery. *Sport Traumatol Arthrosc* 15: 9–16, 2007.
106. Rozzi SL, Lephart SM, Gear WS, and Fu FH. Knee joint laxity and neuromuscular characteristics of male and female soccer and basketball players. *Am J Sports Med* 27: 312–319, 1999.
107. Schiltz M, Lehance C, Maquet D, Bury T, Crielaard JM, and Croisier JL. Explosive strength imbalances in professional basketball players. *J Athl Train* 44: 39–47, 2009.
108. Schmitz RJ, Shultz SJ, and Nguyen AD. Dynamic valgus alignment and functional strength in males and females during maturation. *J Athl Train* 44: 26–32, 2009.
109. Small K, McNaughton LR, Greig M, Lohkamp M, and Lovell R. Soccer fatigue, sprinting and hamstring injury risk. *Int J Sports Med* 30: 573–578, 2009.
110. Soligard T, Myklebust G, Steffen K, Holme I, Silvers H, and Bizzini M, et al. Comprehensive warm-up programme to prevent injuries in young female footballers: Cluster randomised controlled trial. *BMJ* 337: a2469, 2008.
111. Solomonow M and Krogsgaard M. Sensorimotor control of knee stability. A review. *Scand J Med Sci Sports* 11: 64–80, 2001.
112. Sport & Fitness Industry Association. *Sport & Fitness Industry Association. Sports, Fitness, and Leisure Activities Topline Participation Report*, 2013. Available at: https://www.sfia.org/reports/301_2013-Sports%2C-Fitness%2C-and-Leisure-Activities-Topline-Participation-Report.
113. Sugimoto D, Myer GD, Bush HM, and Hewett TE. Effects of compliance on trunk and hip integrative neuromuscular training on hip abductor strength in female athletes. *J Strength Cond Res* 28: 1187–1194, 2014.
114. Sugimoto D, Myer GD, Bush HM, Klugman MF, Medina McKeon JM, and Hewett TE. Compliance with neuromuscular training and anterior cruciate ligament injury risk reduction in female athletes: A meta-analysis. *J Athl Train* 47: 714–723, 2012.
115. Sugimoto D, Myer GD, Foss KDB, and Hewett TE. Dosage effects of neuromuscular training intervention to reduce anterior cruciate ligament injuries in female athletes: Meta- and sub-group analyses. *Sports Med* 44: 551–562, 2014.
116. Sugimoto D, Myer GD, McKeon JM, and Hewett TE. Evaluation of the effectiveness of neuromuscular training to reduce anterior cruciate ligament injury in female athletes: A critical review of relative risk reduction and numbers-needed-to-treat analyses. *Br J Sports Med* 46: 979–988, 2012.
117. Thorborg K, Couppé C, Petersen J, Magnusson SP, and Hölmich P. Eccentric hip adduction and abduction strength in elite soccer players and matched controls: A cross-sectional study. *Br J Sports Med* 45: 10–13, 2011.

118. Tropp H, Ekstrand J, and Gillquist J. Stabliometry in functional instability of the ankle and its value in predicting injury. *Med Sci Sports Exerc* 16: 64–66, 1984.
119. Tveit M, Rosengren BE, Nilsson JA, and Karlsson MK. Former male elite athletes have a higher Prevalence of osteoarthritis and arthroplasty in the hip and knee than expected. *Am J Sports Med* 40: 527–533, 2012.
120. Van der Sluis A, Elferink-Gemser MT, Brink MS, and Visscher C. Importance of peak height velocity timing in terms of injuries in talented soccer players. *Int J Sports Med* 36: 327–332, 2015.
121. Van der Sluis A, Elferink-Gemser MT, Coelho-e-Silva MJ, Nijboer JA, Brink MS, and Visscher C. Sport injuries aligned to peak height velocity in talented pubertal soccer players. *Int J Sports Med* 35: 351–355, 2014.
122. Verkhoshansky Y. *Supertraining* (6th ed). Rome, Italy: Verkhoshansky, 2009.
123. Willson JD, Kernozek TW, Arndt RL, Reznicek DA, and Scott Straker J. Gluteal muscle activation during running in females with and without patellofemoral pain syndrome. *Clin Biomech* 26: 735–740, 2011.
124. Witvrouw E, Lysens R, Bellemans J, Cambier D, and Vanderstraeten G. Intrinsic risk factors for the development of anterior knee pain in an athletic population. A two-year prospective study. *Am J Sports Med* 28: 480–489, 2000.
125. Wojtys EM, Huston LJ, Schock HJ, Boylan JP, and Ashton-Miller JA. Gender differences in muscular protection of the knee in torsion in size-matched athletes. *J Bone Joint Surg Am* 85: 782–789, 2003.
126. Zazulak B, Hewett T, Reeves N, Goldberg B, and Cholewicki J. Deficits in neuromuscular control of the trunk predict knee injury risk: A prospective biomechanical-epidemiologic study. *Am J Sports Med* 35: 1123–1130, 2007.
127. Zazulak B, Ponce P, and Straub S. Gender comparison of hip muscle activity during single-leg landing. *J Orthop Sports Phys Ther* 35: 292–299, 2005.

CSCS professionals apply scientific knowledge to train athletes for peak performance.

CSCS is the ONLY accredited strength and conditioning certification.

Being certified by the NSCA means being certified by the best.

The most sought-after strength and conditioning coaches have their CSCS credential.

CSCS
NATIONAL STRENGTH & CONDITIONING ASSOCIATION
Est. 1985
SPECIALIST

REACH HIGHER IN YOUR CAREER»»»

NSCA
NATIONAL STRENGTH AND
CONDITIONING ASSOCIATION

everyone **stronger**
NSCA.com